

Technology Stack

Date	25 May 2026
Team ID	xxxxxx
Project Name	Smart Lender – AI Loan Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, Bootstrap</i>	<i>Creates a responsive and user-friendly interface.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Handles application logic and ML model integration.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset</i>	<i>Stores applicant data for model training and testing.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Flask Local Server</i>	<i>Enables local deployment and application testing.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Manages source code and supports collaboration.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, Pandas, NumPy, XGBoost, Matplotlib, Seaborn</i>	<i>Used for data preprocessing, visualization, machine learning, and model evaluation.</i>